MIL-E-1/1317D
2 February 1976
SUPERSEDING
MIL-E-1/1317C
12 April 1974

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 3B29W

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Diode, high-vacuum

See figure 1

Mounting position: Any

Weight: 3 ounces (85 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ib	Ib
Unit:	Vac	kv	a	mAdc
Shunt diode: See note 1				
Maximum:	2.75	10.0	8.0	18
Minimum:	2.25	---	---	---
Rectifier:				
Maximum:	2.75	16.0	0.25	65
Minimum:	2.25	---	---	---
TEST CONDITIONS:	2.5	---	---	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1231	Peak emission	-	eb = 4,000 v	0.65	II	is	8.0	---	a
1256	Electrode current (anode)	-	Eb = 130 Vdc	0.65	II	Ib	90	---	mAdc
1301	Heater current	-		0.65	II	If	4.4	5.4	Aac
1353	Operation of rectifiers	7	epx = 16 kv; Ib = 70 mAdc; ib = 275 ma	0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1101	Secureness of base, cap, or insert	-		---	---	---	20	---	in./lb
1027	Temperature cycling	-	-50°to +85°C; one cycle	---	---	---	---	---	---
1331	Direct-interelectrode capacitance	-		---	---	Cpk	2.0	3.0	pF
1041	Shock	-	Accel = 375 G	---	---	---	---	---	---
1031	Sweep-frequency vibration	-		---	---	---	---	---	---
---	Shock and sweep-frequency vibration end points:								
1231	Peak emission	-		---	---	is	8.0	---	a
1353	Operation of rectifiers	-		---	---	---	---	---	---
---	Operation of shunt diode	3		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test (1)	4	Group C; t = 500 hours	---	---	---	---	---	---
---	Life-test (1) end point:								
1256	Electrode current (anode)	-		---	---	Ib	85	---	mAdc
---	Life test (2)	5	Group D	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Life-test (2) end point:								
1231	Peak emission	-		---	---	is	8.0	---	a
1031	Vibration-fatigue test	6	Ef = 2.5 Vac	---	---	---	---	---	---
---	Vibration-fatigue test end points:								
1231	Peak emission	-		---	---	is	8.0	---	a
1353	Operation of rectifiers	-		---	---	---	---	---	---
---	Mechanical resonance	2,6		---	---	---	---	---	---

NOTES:

1. During any microsecond interval, pulse duration should not exceed 25 µs.
2. The fundamental resonant frequency of all cold parts within the tube shall not be between 10 and 165 Hz. In performing the test, the tube shall be vibrated with sinusoidal motion.
3. Operate tube as a shunt diode in a thyratron modulator with the following circuit conditions: t_p = 1.0 µs, 50 ohms; load resistor = 0 ohm; shunt diode series resistor = 1,000 ohms; peak thyratron forward voltage = 10 kv; repetition rate = 280 Hz. There shall be no sign of arcing or gas during the last 2 minutes of a test interval not to exceed 5 minutes.
4. Rectifier circuit, epx = 10 kv, Ib = 100 mAdc, ib = 300 ma.
5. The tube shall be operated under the conditions specified in the first sentence of note 3, 1 minute "on", 3 minutes "off", for a total "on" time of 50 hours. There shall be no sign of arcing or gas during the test.
6. (D) This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.
7. (D) This test is to be the first test performed at the conclusion of the holding period.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 17, 80
DSA - ES

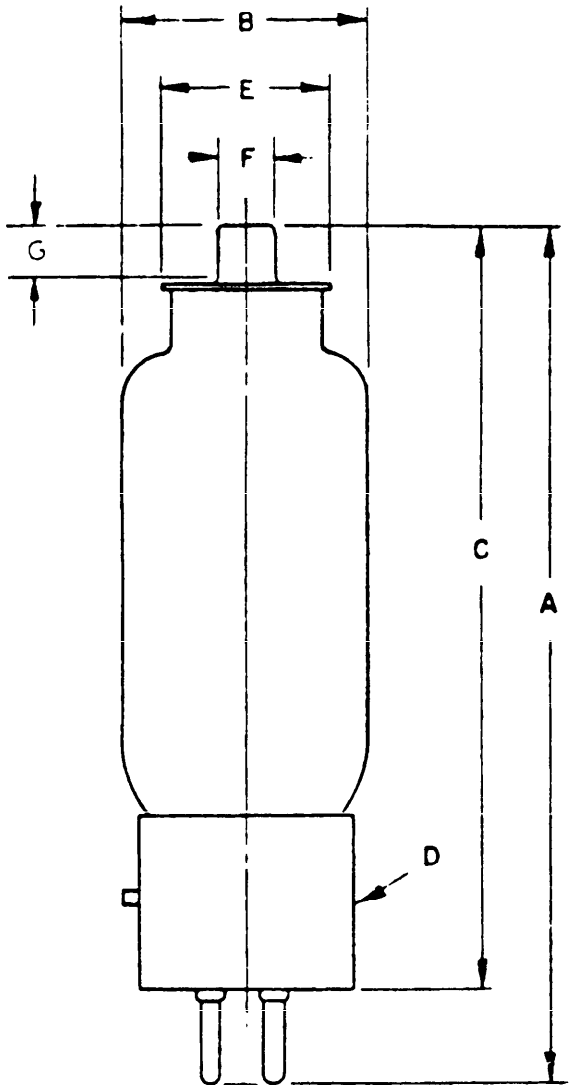
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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PIN CONNECTIONS	
Pin No.	Element
1	h
2	nc
3	nc
4	h, k
Cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	4.875 (123.83)	5.375 (136.53)
B	1.438 (36.53)	1.562 (39.67)
C	4.250 (107.95)	4.750 (120.65)
Quality conformance inspection, part 3 (see note)		
D	Base: A4-10 (EIA)	
E	.844 (21.44)	.906 (23.01)
F	.355 (9.02)	.365 (9.27)
Reference dimensions		
G	.375 (9.53)	

D

NOTE: Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 3B29W.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

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